Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
MCA | Sem-4 | Summer-2023

Course Code	:	2105CS412	**Date**	:	07-04-2023
Course Name	:	Computer Networks	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define Network Topology. Explain topologies with a diagram.	**4**
	(B)	Define types of computer networks. (LAN, MAN, WAN).	**3**
		OR	
		List differences between circuit switching & packet switching.	
	(C)	Explain the OSI Reference Model.	**7**
		OR	
		Explain the TCP/IP Reference Model.	
Q.2	**(A)**	Compare switch and router.	**4**
	(B)	Explain the twisted-pair cable in detail.	**3**
		OR	
		Explain the co-axial cable in detail.	
	(C)	Explain radio wave in detail.	**7**
		OR	
		Discuss about microwave transmission in detail.	
Q.3	**(A)**	Explain CRC method with example.	**4**
	(B)	Explain flow control with example.	**3**
		OR	
		Describe congestion control approach.	
	(C)	Explain Go back N protocol with diagram.	**7**
		OR	
		Explain selective repeat protocol with diagram.	

Q.4	**(A)**	Given IP address 169.25.45.40/20 Calculate below details: a) Which Class? b) Subnet mask? c) What are the total subnets? d) What are the total hosts?	**4**
	(B)	Explain IPv4 datagram format.	**3**
		OR	
		Compare IPv4 and IPv6.	
	(C)	Explain distance vector routing algorithm with suitable example.	**7**
		OR	
		Describe link state routing algorithm with suitable example.	
Q.5	**(A)**	Compare TCP and UDP.	**4**
	(B)	Explain DNS with example.	**3**
		OR	
		Describe SMTP with example.	
	(C)	Explain RDT 2.0 with diagram.	**7**
		OR	
		Describe multiplexing in transport layer.	

***